Experiment - 5

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Experiment 5: Average Fuel Efficiency and Percentage Improvement

Aim:

To calculate the average fuel efficiency of car models and determine the percentage improvement between two models using NumPy.

Algorithm:

1. Import NumPy.
2. Create an array containing the fuel efficiency of car models.
3. Calculate the average using np.mean().
4. Calculate the percentage improvement between two models.
5. Display the results.

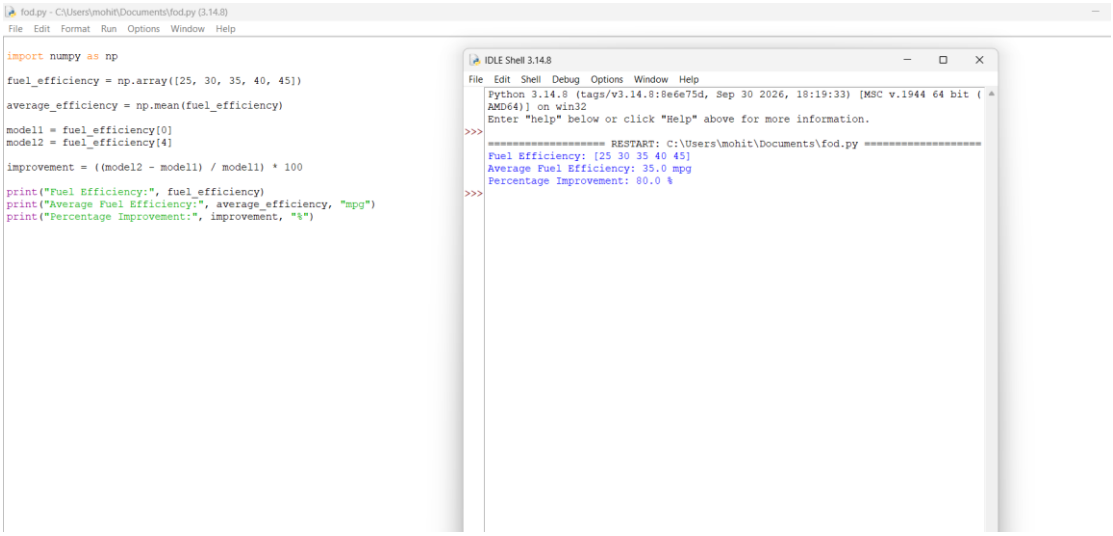
Python Code (IDLE):

```
import numpy as np
fuel_efficiency = np.array([25, 30, 35, 40, 45])
average_efficiency = np.mean(fuel_efficiency)
model1 = fuel_efficiency[0]
model2 = fuel_efficiency[4]
improvement = ((model2 - model1) / model1) * 100
print("Fuel Efficiency:", fuel_efficiency)
print("Average Fuel Efficiency:", average_efficiency, "mpg")
print("Percentage Improvement:", improvement, "%")
```

Input:

Fuel efficiency = [25, 30, 35, 40, 45]

Output:



Result:

Thus, the average fuel efficiency was 35 mpg, and Model 2 (45 mpg) showed an 80% improvement over Model 1 (25 mpg).